A Functionally Enriched Form of DAP as an EarthCube Building Block

An NSF-funded project dubbed ODSIP
(Open Data-Service Invocation Protocol)

D Fulker, PI, with Co-PIs
B Blanton, S Businger
& P Cornillon

by Dave Fulker
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FOUNDATIONAL CONCEPTS

from DODS (now OpenDAP) circa 1994

- URL ≈ dataset | URL <u>w/ constraint</u> ≈ subset
- Retrieve →
  - *dataset descriptions (metadata)*
  - *dataset content (typed/structured)*
- Retrieval methods are built into analysis apps via (multilingual) <u>libraries</u>
  - → *many, diverse clients*
  - → *flexible data use*

DAP has long enabled several <u>types</u> of

Subset Creation

- Select variables by name
- Select table rows (instances of a "sequence") by value constraints
- Select subarrays by limiting the ranges of (named) indices
- *But*, for example:
  - ►*Value constraints can't apply to arrays*
  - ►*Arrays can't be extracted as sequences*
  - ►...

2011-2014, thanks to NOAA funding,

OPENDAP-UNIDATA LINKED SERVERS (OPULS)

Aligning & Linking Open-Source Software for Web-Based Scientific Data Exchange

- DAP4 (1st major DAP2 revision in 20 yrs): "increase OpenDAP/Unidata conformance"
- Two demonstrations
  - *Server-side subsetting of non-rectangular meshes*
  - *Modes of asynchronous access*

IDEAS FOR ODSIP

- **Stimulated by OPULS successes—**
  - ***(Beyond DAP4)***
  - ***Subsetting of non-rectangular meshes***
  - ***Experiments with asynchronous access***
- **—the question arises…**

→ ***Why not support invocation of <u>additional</u> server-side functions?***

Some examples of largely unmet

Data-Retrieval Needs

- Richer subsetting
  - *Non-rectangular structures*
    - Triangular/polygonal meshes
  - *Value constraints on arrays*
    - What shape result?
  - *Other constraints requiring calculations…*

- Additional ops* (pre-retrieval!)
  - *Statistical summarization, binning…*
  - *Remapping, regridding…*
  - *Feature extraction…*

*Note: As a rule, new ops should <u>decrease</u> data-xfer volumes

DAP4* *sets stage for* ODSIP**

(Open Data-Service Invocation Protocol)

- OpenDAP-Unidata alignment
- Clearer, crisper descriptions:
  - *DAP4 data model*
  - *DAP4 queries & responses*
  - *Web-services context (modernized)*
- Hooks for extensibility, including new server functions
  - *Prototype: subsetting UGRIDS*

*Thanks to NOAA & **NSF

Under DAP4 (as with DAP2)

URL ≈ DATASET (GRANULE)

http://laboratory.edu/device/experiment/granule.dmr

Domain name often is an organization's web server.	Servers often have hierarchical collections.	Each URL references a distinct DAP "dataset."	Suffixes specify return types.
Depending on the suffix, DAP returns (textual) metadata or (binary) content, with options for human- and machine-readable forms (XML, NetCDF4...). Suffix "dmr" yields metadata only.			

DAP4

URL+QUERY ≈ SUBSET *or…*

http:/…/granule.nc4?dap4.ce=constraints&dap4.func=functions

Dataset identifier as above, except return-type is NetCDF4 (= HDF)	DAP4 "constraint expressions" yield sub-arrays & other proper subsets	DAP4 "function expressions" enable (server-specific) extensions
Constraints specify subsets by array indices, variable names, and (in some cases) content. Function expressions allow new services (statistics, etc.) to be offered as part of data retrieval.		

The &dap4.fun=… expression enables DAP extensions such as server functions

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PI: Dave Fulker (OpenDAP)
Co-Investigators: Brian Blanton, Steven Businger, Peter Cornillon, James Gallagher, Mohan Ramamurthy

OpenDAP concept for an

EARTHCUBE BUILDING BLOCK

Data Acquisition as a Web Service with Extensive Pre-Retrieval Operations

DAP4 as Foundation

ODSIP Is a Prototyping Effort

Needed (pre-retrieval*) ops

- *Statistical summarization*
  - *Calculate* ***means****; form* ***masks****; accumulate counts (****binning****)…*
- *Regridding/resampling*
  - *Remap onto* ***spec'd meshes****…*
- Criteria-driven subset creation
  - *Select/build* ***data structures*** *that* ***satisfy a predicate***
- *Feature extraction*
  - *Construct* ***lines/polygons*** *from images, grids…*

ODSIP project outcomes

- A new data-acquisition method with built-in pre-retrieval functions
- Prototype usage demonstrating data-xfer economies in 3 geo contexts
  - ***Dynamic downscaling*** *- climate predictions for native-Hawaiian use (local scales)*
  - ***Storm surge prediction*** *- for emergency management in coastal NC*
  - ***SST-front analysis/synthesis*** *- for use in chlorophyl studies*
- An effective building-block concept, meeting diverse EarthCube needs

IN CONCLUSION, THE
ODSIP PROJECT WILL

- **Develop a service-invocation protocol that**
  - *Extends the well-used OpenDAP protocol*
  - *Supports rich pre-retrieval processing*
- **Prototype its use in 3 demanding geo contexts**
  - *Climate-model downscaling for native-Hawaiian use*
  - *Storm surge prediction for coastal NC*
  - *SST front analysis/synthesis from satellite imagery*
- **Engage the EarthCube community…**